Calibrated Dependence-Aware Day-Ahead Bidding of Virtual Power Plants Under Joint Renewable, Load, and Price Uncertainty

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Abstract

Virtual power plants (VPPs) aggregate renewable generation, flexible demand, energy storage, and other distributed resources whose joint uncertainty directly affects the economic value and physical deliverability of market commitments. Existing stochastic and risk-aware VPP bidding models have predominantly relied on independent or simplified uncertainty representations, Gaussian dependence structures, robust uncertainty sets, or scenario-generation procedures that are evaluated primarily from a statistical rather than a downstream decision perspective. This paper develops a calibrated dependence-aware stochastic bidding framework for VPP participation in sequential electricity markets. The proposed framework combines pre-decision forecast calibration, conformal uncertainty calibration, a Student-t copula for joint dependence and tail behavior, tail-preserving scenario reduction, and a rolling-horizon stochastic mixed-integer linear programming (MILP) model with seven operational stages. The optimization explicitly represents day-ahead commitments, sequential flexibility carry-over, renewable dispatch, battery operation, demand-response flexibility, imbalance settlement, and N−1 surviving flexible capability. A linear CVaR formulation is incorporated to quantify and control lower-tail economic risk. The empirical framework is designed around German electricity-market data from SMARD and uses strict chronological information-set rules to prevent look-ahead bias. The probabilistic engine generates 5,000 joint trajectories and evaluates reduced scenario sets of different cardinalities. Preliminary validation confirms strong temporal-dependence preservation, with a pooled mean ACF error of 0.00394 and maximum ACF error of 0.01873, while the reduced scenario sets preserve probability mass exactly. The repaired MILP formulation has also passed independent numerical smoke testing using the open-source HiGHS solver through SciPy. Final economic superiority claims are intentionally withheld until the common out-of-sample settlement experiment is completed. The resulting framework is intended to provide an auditable link between probabilistic forecasting quality, tail dependence, stochastic bidding decisions, and realized VPP performance.

Index Terms— Virtual power plant, stochastic MILP, day-ahead bidding, Student-t copula, conformal calibration, CVaR, scenario reduction, demand response, battery energy storage, uncertainty, out-of-sample validation.

Nomenclature

Abbreviation	Description
Abbreviations	
CVaR	Conditional Value-at-Risk
DER	Distributed Energy Resource
DG	Distributed Generation

DR	Demand Response
DRP	Demand Response Provider
BESS	Battery Energy Storage System
DA	Day-Ahead
ID	Intraday
RM	Reserve Market
OOS	Out-of-Sample
MILP	Mixed-Integer Linear Programming
PV	Photovoltaic
VPP	Virtual Power Plant
WPP	Wind Power Plant
PIT	Probability Integral Transformation
ACF	Autocorrelation Function
AIC	Akaike Information Criterion
DRO	Distributionally Robust Optimization
Sets and Indices	
$\omega \in \Omega$	Scenario index
$t \in \mathcal{T}$	Time index (hourly resolution)
$s \in \mathcal{S}$	Operational stage index ($\mathcal{S} = \{1, \dots, 7\}$)
$g \in \mathcal{G}$	Distributed generation unit index
$b \in \mathcal{B}$	Battery energy storage system index
$p \in \mathcal{P}$	Photovoltaic unit index
$w \in \mathcal{W}$	Wind power plant index
$d \in \mathcal{D}$	Dispatch group index for N−1 constraints
$j \in \mathcal{J}$	Demand response group index
Parameters	
$P_{j,t}^{DR}$	Power usage of the j-th flexible demand group at time t(MW)
$\eta_b^{ch}, \eta_b^{dch}$	Charging and discharging efficiency factors of BESS b
λ_t^{DA}	Day-ahead market price at time t($/MWh)
$\lambda_{t,\omega}^{RT}$	Real-time market price in scenario ωat time t($/MWh)
$\lambda_{j,\omega}^{DR}$	Price offered to demand response group $j$in scenario ω($/MWh)
$\lambda_{j,t}^{DR}$	Time-varying DR price for group $j$at time t($/MWh)
$\lambda_{b,t}^{ESS}$	Operational cost of BESS $b$at time t($/MWh)
α	Confidence level for CVaR (e.g., 0.95)
β	Risk-aversion coefficient (weight on CVaR in objective)
π_ω	Probability of scenario ωoccurrence
$C_{g,t}^{DG}$	Marginal cost of distributed generation unit $g$at time t($/MWh)
E_b^{min}, E_b^{max}	Minimum and maximum state-of-charge limits for BESS b(MWh)
$P_b^{ch,max}, P_b^{dch,max}$	Maximum charging and discharging power of BESS b(MW)
$P_{w,t}^{W,avail}$	Available wind power from plant $w$at time t(MW)
$P_{p,t}^{PV,avail}$	Available PV power from unit $p$at time t(MW)
DR_j^{max}	Maximum demand response capacity of group j(MW)
$E^{DR,max}$	Maximum cumulative DR energy over the response window (MWh)
c_t^{DR}	Unit cost of demand response activation at time t($/MWh)
$P_t^{DA,comm}$	Day-ahead committed power (MW)
F^{avail}	Total available flexibility capacity (MW)
F_r^{comm}	Flexibility committed at stage r(MW)
$F_{d,t,\omega}^{surv}$	Surviving flexible capacity in dispatch group $d$after N−1 contingency (MW)
$R_{d,t}^{comm}$	Reserve commitment of dispatch group $d$at time t(MW)
M	Big-M constant for logical constraints
R	Correlation/dependence matrix of Student-t copula
ν	Degrees-of-freedom parameter of Student-t copula
Decision Variables	
x_s	Decision vector at operational stage s
x_s^*	Optimal decision at stage s(fixed for subsequent stages)
$P_{t,\omega}^{PV}$	Dispatched PV power in scenario ωat time t(MW)
$P_{t,\omega}^{W}$	Dispatched wind power in scenario ωat time t(MW)
$P_{t,\omega}^{DG}$	Dispatched DG power in scenario ωat time t(MW)
$P_{t,\omega}^{ch}, P_{t,\omega}^{dch}$	Charging and discharging power of BESS in scenario ωat time t(MW)
$E_{t,\omega}$	State of charge of BESS in scenario ωat time t(MWh)
$P_{t,\omega}^{grid}$	Net grid exchange power in scenario ωat time t(MW)
$DR_{t,\omega}$	Activated demand response in scenario ωat time t(MW)
$L_{t,\omega}$	Load in scenario ωat time t(MW)

$I^{+}_{t,\omega}, I^{-}_{t,\omega}$	Positive and negative imbalance in scenario ωat time t(MW)
$z_{t,\omega}$	Binary variable preventing simultaneous BESS charge/discharge
Π_{ω}	Operating profit in scenario ω($)
L_{ω}	Loss in scenario ω($) (negative of profit)
η	Auxiliary CVaR variable (VaR threshold)
ξ_{ω}	Auxiliary CVaR variable for excess loss in scenario ω($)
R^{DA}_{ω}	Day-ahead market revenue in scenario ω($)
R^{ID}_{ω}	Intraday market revenue in scenario ω($)
R^{RT}_{ω}	Real-time market revenue in scenario ω($)
R^{DR}_{ω}	Demand response revenue in scenario ω($)
R^{RES}_{ω}	Reserve market revenue in scenario ω($)
C^{DG}_{ω}	Distributed generation cost in scenario ω($)
C^{BESS}_{ω}	Battery operational cost in scenario ω($)
C^{DR}_{ω}	Demand response activation cost in scenario ω($)
C^{imb}_{ω}	Imbalance settlement cost in scenario ω($)
F^{rem}_{s}	Remaining flexibility after stage s(MW)
e^{x}_{t}	Forecast error for variable $x$at time t(MW or $/MWh)
$\tilde{e}^{x}_{t}$	Calibrated forecast error for variable $x$at time t(MW or $/MWh)
u^{x}_{t}	Probability integral transform (PIT) value for variable $x$at time t
$z_{\omega,t}$	Latent vector from Student-t copula in scenario ωat time t
$e^{x}_{\omega,t}$	Generated forecast-error scenario for variable x
$x_{\omega,t}$	Generated physical scenario for variable x
$\rho_{x}(l)$	Autocorrelation of error process $x$at lag l
Operational Stage Definitions	
$s = 1$	Day-ahead (DA) bid submission
$s = 2$	Intraday (ID) adjustment 1
$s = 3$	Intraday (ID) adjustment 2
$s = 4$	Real-time (RT) pre-dispatch
$s = 5$	Balancing market activation
$s = 6$	Final settlement
$s = 7$	Post-event evaluation / OOS reconciliation

I. INTRODUCTION

The increasing penetration of renewable distributed energy resources has transformed the operational role of virtual power plants (VPPs). By aggregating photovoltaic (PV) generation, wind generation, battery energy storage systems (BESSs), flexible demand, and other distributed resources, a VPP can participate in electricity markets as an aggregated market-facing entity. This aggregation creates economic opportunities but simultaneously exposes the VPP to forecast uncertainty, price volatility, imbalance costs, intertemporal resource constraints, and uncertainty in the availability of flexible capacity.

The resulting bidding problem is fundamentally different from deterministic scheduling. A day-ahead energy position that is profitable under a point forecast may become economically inferior when renewable forecast errors, load deviations, and market-price shocks occur jointly. Similarly, allocating excessive flexibility to the energy market may reduce the VPP's ability to satisfy later reserve or balancing obligations. Recent literature has therefore investigated stochastic

programming, robust optimization, CVaR-based bidding, chance constraints, and distributionally robust optimization for VPP participation in energy and ancillary-service markets.

Several recent studies have strengthened this research direction. Nemati et al. investigated flexible robust bidding of renewable VPPs in sequential markets under asymmetric uncertainties [1], while Lu et al. developed a stochastic VPP scheduling model accounting for price perturbations [2]. Vahedipour-Dahraie et al. considered risk-averse energy and reserve scheduling with demand response [3]. More recent studies have considered distributionally robust bidding [4], multi-market stochastic bidding [5], and CVaR-oriented formulations [6]. These developments demonstrate that uncertainty and economic risk are no longer peripheral considerations in VPP optimization.

Nevertheless, a critical methodological issue remains. The quality of a stochastic bidding model depends not only on the optimization formulation but also on how the uncertainty distribution is constructed. Forecast errors in renewable generation, load, and electricity prices are neither necessarily independent nor adequately characterized by light-tailed Gaussian dependence. A VPP may simultaneously experience high renewable underproduction, elevated load, and unfavorable market prices. Such compound events are particularly relevant for tail-risk-aware bidding. The project's probabilistic architecture therefore models the joint forecast-error vector rather than perturbing each variable independently.

More recent work has extended stochastic VPP models to ancillary-service participation, demand response, storage, and multiple markets. Wang et al. investigated stochastic bidding for VPP-enabled ancillary services [7], while Mujeeb et al. considered risk-averse two-stage stochastic bidding in energy and frequency-regulation markets [8].

However, most stochastic formulations remain strongly dependent on the assumed scenario-generation mechanism. If scenarios do not preserve the dependence and tail characteristics of the underlying forecast errors, a mathematically correct stochastic program can still produce economically misleading bids.

II. Literature Review

A. VPP Bidding under Uncertainty

Virtual power plants aggregate distributed generation, storage, and flexible demand so that they can be offered into wholesale electricity markets as a single dispatchable entity. Much of the recent

literature on VPP day-ahead (DA) bidding formulates the problem as a two-stage stochastic program, in which the first-stage DA bid is fixed before uncertainty is revealed and a second-stage recourse action absorbs the resulting imbalance. Representative recent examples include a two-stage scenario-oriented model for a multi-energy VPP bidding jointly into the day-ahead energy market and the frequency-regulation reserve market [9], a bidding strategy that explicitly represents the VPP's physical–economic feasible operation region when participating in joint energy and reserve markets [11], and a game-theoretic, Stackelberg-based day-ahead bidding strategy for a VPP aggregating heterogeneous distributed resources under a CVaR risk criterion [12]. Learning-based alternatives have also been explored; for example, a price-maker VPP bidding strategy trained with a multi-agent twin-delayed deep deterministic policy gradient algorithm [13]. Across this body of work, the two-stage MILP remains the dominant and most tractable structure, which is why it is retained as the backbone of the model proposed in this paper.

B. Risk Measures: CVaR and Distributionally Robust Optimization

Because a risk-neutral expected-profit objective can expose a VPP operator to large losses in adverse scenarios, Conditional Value-at-Risk (CVaR) has become the standard risk measure in this literature, valued for its convexity and its exact linear reformulation via auxiliary variables. A recent and comprehensive review categorizes VPP risk-averse bidding techniques into stochastic programming with CVaR, robust optimization, distributionally robust optimization (DRO), and emerging game-theoretic and learning-based approaches, and observes that CVaR-based stochastic programming remains the most widely adopted method in practice [10]. CVaR has been embedded directly into VPP bidding formulations that include mobile or vehicle-based storage assets, feasible-region constraints for vehicle-to-grid participation [9], and bi-level formulations coupling a VPP with power-to-hydrogen conversion [10]. Outside the VPP literature strictly defined, CVaR has also been used to manage the daily and hourly risk of an electricity retailer engaged in virtual bidding with battery storage [15], a formulation that is structurally close to the VPP bidding problem addressed here. A parallel and increasingly popular direction replaces the fixed-distribution assumption of stochastic programming with distributionally robust optimization: a Wasserstein-distance-based DRO/chance-constrained bidding model has been proposed for VPPs participating in coupled electricity–carbon markets [14], hedging against ambiguity in the underlying probability distribution rather than committing to a single empirical one. DRO and CVaR are complementary rather than competing: DRO addresses ambiguity in the distribution itself, while CVaR addresses the shape of the loss tail once a distribution (or ambiguity set) has been fixed. The present paper retains CVaR as the risk measure of choice for its computational tractability inside a MILP, while addressing distributional risk through calibrated, heavy-tailed dependence modeling rather than through an explicit ambiguity set.

C. Dependence Modeling and the Gaussian-Copula Default

A separate and, for this paper, central strand of the literature concerns how joint uncertainty across multiple correlated random variables — renewable output, load, and price — is represented for scenario-based optimization. Early work modeled the joint distribution of forecast errors across multiple wind farms using a Gaussian mixture model to capture non-Gaussian, interdependent behavior beyond what a single Gaussian assumption allows [19]. More broadly, copula-based approaches decouple the marginal distribution of each uncertain variable from the dependence

structure that links them, and have been applied to spatially correlated wind generation, wind–solar joint scenario generation, and price–load dependence. In nearly all of these applications, however, the Gaussian copula is the default choice because of its analytical convenience and ease of sampling, even though it is well known to underrepresent joint tail events — that is, the simultaneous occurrence of extreme outcomes such as a severe renewable shortfall coinciding with an extreme price spike. This is precisely the behavior that a heavy-tailed dependence structure, such as the Student-t copula, is designed to capture, since its degrees-of-freedom parameter directly controls the strength of joint tail dependence. Despite this well-established property in the statistics and financial-risk literature, its explicit, calibrated use inside a VPP day-ahead bidding MILP — jointly with a CVaR objective and validated on real market data — has not been demonstrated in the papers reviewed for this study.

D. Calibration and Conformal Prediction

A further limitation of most stochastic VPP bidding formulations is that the marginal distributions feeding the scenario generator are assumed to be known or are fit parametrically without any distribution-free calibration guarantee. Conformal prediction offers a model-agnostic alternative: it converts point or quantile forecasts into prediction intervals with a finite-sample coverage guarantee, without requiring a parametric error distribution. Conformal methods have recently been applied to day-ahead and real-time balancing-market electricity price forecasting, combining ensemble batch prediction intervals with sequential predictive conformal inference to obtain calibrated, reliable intervals under distribution shift [20], [21]. To date this line of work has remained focused on forecasting and has not been integrated into the scenario-generation and optimization pipeline of a VPP bidding model — which is one of the elements the present paper incorporates through conformal calibration of the marginal forecast-error distributions prior to copula fitting.

E. Storage Degradation and Deliverable Reserve

Two further modeling details recur across the literature and are treated explicitly in the proposed formulation. First, several VPP bidding models retain battery degradation only as a simplified linear throughput cost, or omit it altogether, even where the aggregated portfolio includes vehicle-to-grid or mobile storage assets whose cycling behavior materially affects lifetime cost [16], [17]. Second, reserve or flexibility offers are frequently sized from the nominal aggregated capacity of the VPP's assets rather than from the capacity that remains deliverable after the loss of any single asset, which can overstate the reserve that a VPP can reliably provide under contingency.

F. Research Gap

Taken together, the reviewed literature shows that

(i) CVaR-based two-stage stochastic bidding is well established for VPPs [9], [11], [12], [17], [18];

(ii) DRO provides an alternative treatment of distributional ambiguity but has not been combined with an explicitly calibrated, heavy-tailed dependence structure [14];

(iii) copula-based dependence modeling for renewable and price uncertainty exists but defaults overwhelmingly to Gaussian or non-parametric mixture assumptions that underrepresent joint tail risk [19]; and

(iv) conformal calibration has been validated for electricity-price forecasting in isolation but not linked to scenario generation for market bidding [20], [21]. No paper identified in this review combines conformal marginal calibration, an explicitly tail-protected Student-t copula, degradation-aware battery cost, and a CVaR-constrained two-stage MILP validated on real day-ahead market data under a strict rolling-origin, no-leakage out-of-sample protocol. This combination constitutes the research gap addressed by the model proposed in Section III–IV of this paper.

Comparative Positioning Table

Comparison of representative studies against the model proposed in this paper. "—" indicates the element is not part of the cited study's formulation, based on the abstract/full text reviewed.

Study	Uncertainty Representation	Dependence Model	Risk Measure	BESS Degradation	Real Market Data	Deliverable (N-1) Reserve
Mujeeb et al., *Int. Trans. Electr. Energy Syst.*, 2025 [9]	Scenario-based, two-stage stochastic	Independent scenarios	CVaR	—	—	—
Fan & Liu, *IET Renew. Power Gener.*, 2024 [14]	Wasserstein ambiguity set (DRO)	— (moment/metric-based, not copula)	Chance-constrained (DRO)	—	—	—
Wang et al., *Appl. Energy*, 2018 [19]	Conditional, non-Gaussian scenario generation	Gaussian mixture model	— (forecasting only, no market bidding)	—	Real wind data	—
O'Connor et al., *Energy and AI*, 2025 [20]	Conformal prediction intervals	—	— (forecasting only, no market bidding)	—	Real DA/balancing price data (Ireland)	—
Yoo & Kim, arXiv:2503.05366, 2025 [18]	Bi-level scenario-based	Independent scenarios	CVaR	—	—	—
Proposed model (this paper)	Conformal-calibrated marginals + joint scenarios	Student-t copula with explicit tail protection	CVaR (linearized, two-stage MILP)	Yes (throughput/DoD-aware, linearized)	Real day-ahead market data, rolling-origin OOS	Yes (dispatch-group, N-1 deliverability)

III. System Description and Modeling Assumptions

The virtual power plant (VPP) considered in this paper aggregates photovoltaic (PV) generation, wind generation, a battery energy storage system (BESS), a distributed/dispatchable generation unit, and a flexible demand-response (DR) resource, and participates in the day-ahead (DA) market as its primary product. The formulation is written to also accommodate an optional intraday (ID) and reserve-market (RM) extension, since this is a natural extension of the same asset base, but the core contribution and the experiments in this paper concern DA bidding under joint renewable–price dependence.

The following assumptions are made explicit so that every constraint below can be checked against them:

1. Forecast errors for PV, wind, load, and DA price are treated as stationary within each rolling-origin estimation window (training/calibration/validation), consistent with the walk-forward evaluation protocol described in Section VI.
2. Battery degradation is represented as a linear cost proportional to energy throughput; this is a standard simplification in the MILP literature (see Section II-E) and is stated here as a modeling choice, not a claim of physical accuracy.
3. The VPP is a price-taker: its bid quantity is assumed too small to move the DA clearing price.
4. Imbalance settlement follows a single dual-price penalty structure (positive and negative imbalance priced separately), consistent with the German balancing-market mechanism described in Section II.
5. The optional reserve product, where included, must be deliverable after the loss of any single asset within its dispatch group (N-1), not merely bounded by nominal aggregate capacity.

IV. DATA AND UNCERTAINTY MODELING

A. Data Source and Market Information

The empirical analysis uses publicly available German electricity-market data from SMARD for the period from January 1, 2024, to September 30, 2025. Day-ahead electricity prices are represented at hourly resolution, whereas renewable-generation and load observations are available at 15-min resolution. The physical variables include onshore wind generation, offshore wind generation, photovoltaic generation, actual load, and their corresponding forecast information.

To maintain chronological consistency between forecasting and bidding, the available information is divided into historical training, calibration, validation, and out-of-sample settlement periods. The resulting information structure prevents observations that become available after the bidding decision from entering the corresponding forecast or optimization problem.

The forecast-error variables are defined as

$e_t^x = x_t^{act} - \hat{x_t}, \mathrm{x} \in \{W_{on}, W_{off}, \mathrm{PV}, \mathrm{L}, \mathrm{P}\},$	(1)

Where x_t^{act} is the realized value and $\hat{x_t}$ denotes the corresponding forecast.

The resulting error processes are modeled jointly because simultaneous deviations in renewable generation, load, and market price can have substantially greater economic consequences than isolated forecast errors.

B. Forecast Calibration

The raw forecast errors are first calibrated before probabilistic scenario generation. This step is particularly important for load because the empirical analysis identified a substantial systematic bias in the original load-error series.

For each variable , the calibrated forecast is represented by

$\hat{x}_t^{cal} = \hat{x}_t - \hat{b}_{x,t}$	(2)

where $\hat{b}_{x,t}$ represents the estimated forecast bias.

The calibrated residual is subsequently defined as

$\tilde{e}_t^x = x_t^{act} - \hat{x}_t^{cal}$	(3)

All marginal distributions used by the scenario engine are fitted to $\tilde{e}_t^x$, rather than to the original uncorrected forecast errors.

C. Probability Integral Transformation

The calibrated residuals are transformed into approximately uniform variables using their fitted marginal distributions:

$u_t^x = F_x(\tilde{e}_t^x), \;\; u_t^x \epsilon (0,1)$	(4)

Where $F_x(0)$ is the empirical or fitted marginal CDF of variable x .

The transformed vector is

$u_t = \left[u_t^{PV}, u_t^{W_{on}}, u_t^{W_{off}}, u_t^L, u_t^P\right]^T$	(5)

This transformation separates marginal behavior from cross-variable dependence and provides the input required for copula estimation.

D. Student-t Copula

A Student-t copula is employed to represent nonlinear dependence and joint tail behavior. For uncertain variables,

$C_{v,R}(u_1, \ldots., u_m) = t_{v,R}(t_v^{-1}(u_1), \ldots., t_v^{-1}(u_m))$	(6)

Where R is the dependence matrix and v is the degrees-of-freedom parameter.

The fitted model produced $v = 18.5515$

The validation comparison yielded $AIC_t = -670.7362$

compared with $AIC_{Gaussion} = -670.7362$

The corresponding recorded OOS log-likelihoods were $LL_t = 12.990350$

And $LL_{Gussion} = -30.708711$

These results support the use of the Student-t copula for the subsequent scenario-generation stage.

E. Joint Scenario Generation

A joint latent vector is generated from

$z_{\omega,t} \sim t_v(0, R)$	(7)

and transformed through the Student-t CDF:

$u_{\omega,t}^x = t_v(z_{\omega,t}^x)$	(8)

The inverse marginal transformations then produce forecast-error scenarios:

$e_{\omega,t}^x = F_x^{-1}(u_{\omega,t}^x)$	(9)

The resulting physical scenarios are

$x_{\omega,t} = \hat{x}_t^{cal} + e_{\omega,t}^x$	(10)

A large initial scenario pool is generated before scenario reduction. The previously executed experiment generated 5,000 scenarios; however, that specific run is excluded from the final empirical results because its marginal-fitting implementation used the uncorrected residual columns.

F. Temporal Dependence Validation

The generated scenarios are additionally evaluated against the temporal dependence of the observed error processes. For lag l,

$\rho_x(l) = Corr(e_t^x, e_{t-1}^x)$	(11)

The validation experiment produced a pooled mean absolute ACF error of $MAE_{ACF} = 0.003945$

and a maximum absolute error of $MaxError_{ACF} = 0.018733$

These diagnostics provide an independent check that the generated scenarios reproduce the temporal structure of the calibrated uncertainty processes.

G. Tail-Protected Scenario Reduction

Because solving the complete stochastic MILP using thousands of scenarios is computationally expensive, the generated scenario pool is reduced while preserving representative central and extreme trajectories.

The retained scenario set is

$\Omega_R c\Omega\ , \lvert\Omega_R\rvert = K$	(12)

The principal scenario-count experiments are conducted for

$K\epsilon\{25,50,100,200\}$	(13)

Tail protection prevents extreme scenarios from being eliminated solely because of their low probability. The final selection is evaluated not only using statistical distance but also using decision-level metrics obtained from the OOS optimization.

V. STOCHASTIC MILP FORMULATION

A. Problem Definition

The VPP is modeled as a rolling-horizon stochastic MILP with seven operational stages. The complete horizon is formulated simultaneously for theoretical exposition, whereas operational implementation follows a sequential rolling-horizon procedure.

The model jointly determines energy commitments, flexibility allocation, BESS operation, demand response, real-time adjustments, and risk protection under the retained uncertainty scenarios.

B. Objective Function

The proposed objective maximizes expected operating profit while penalizing downside risk:

$\max \mathbb{E}[\Pi_\omega] - \lambda CVaR(L_\omega)$	(14)

Where $\lambda \geq 0$ is the risk-aversion coefficient and L_ω denotes the scenario loss.

Expected profit is represented by

$\Pi_\omega = R_\omega^{DA} + R_\omega^{ID} + R_\omega^{RT} + R_\omega^{DR} + R_\omega^{RES} - C_\omega^{DG} - C_\omega^{BESS} - C_\omega^{DR} - C_\omega^{imb}$	(15)

C. CVaR Linearization

Following the standard Rockafellar–Uryasev representation,

$CVaR_\alpha(L) = \eta + \frac{1}{1-\alpha}\mathbb{E}[(L-\eta)^+]$	(16)

For each scenario,

$\xi_\omega \geq L_\omega - \eta$	(17)
$\xi_\omega \geq 0$	(18)

Therefore,

$CVaR_\alpha(L) = \eta + \frac{1}{1-\alpha}\sum_{\omega \epsilon \Omega_R} p_\omega \xi_\omega$	(19)

This representation preserves MILP structure.

D. Energy Balance

For each scenario and hour,

$P_{t,\omega}^{PV} + P_{t,\omega}^{W} + P_{t,\omega}^{DG} + P_{t,\omega}^{dis} - P_{t,\omega}^{ch} + P_{t,\omega}^{grid} - DR_{t,\omega} = L_{t,\omega}$	(20)

The equation ensures physical balance between available generation, storage, grid exchange, demand response, and load.

E. Renewable Generation Constraints

For photovoltaic generatiosn,

$0 \leq P_{t,\omega}^{PV} \leq PV_{t,\omega}$	(21)

For wind generation,

$0 \leq P_{t,\omega}^{W} \leq W_{t,\omega}$	(22)

Thus, renewable dispatch cannot exceed scenario-dependent available production.

F. BESS Constraints

The battery energy balance is

$E_{t,\omega} = E_{t-1,\omega} + \eta^{ch} P_{t,\omega}^{ch} - \frac{P_{t,\omega}^{dis}}{\eta^{dis}}$	(23)

The operating limits are

$E^{min} \leq E_{t,\omega} \leq E^{max}$	(24)
$0 \leq P_{t,\omega}^{ch} \leq P^{ch,max} z_{t,\omega}$	(25)
$0 \leq P_{t,\omega}^{dis} \leq P^{dis,max}(1 - z_{t,\omega})$	(26)

where is a binary variable preventing simultaneous charging and discharging.

G. Demand Response

Demand-response activation satisfies

$0 \leq DR_{t,\omega} \leq DR_t^{max}$	(27)

If energy neutrality is imposed over the response window,

$\sum_{t \epsilon T} DR_{t,\omega} \leq E^{DR,max}$	(28)

The corresponding activation cost is

$C_{\omega}^{DR} = \sum_{t} c_t^{DR} DR_{t,\omega}$	(29)

H. Sequential Commitment

The remaining flexibility after each market stage is defined as

$F_s^{rem} = F^{avail} - \sum_{r=1}^{s} F_r^{comm}$	(30)

Consequently,

$F_r^{comm} \leq F_s^{rem}$	(31)

This prevents the same physical flexibility from being sold repeatedly across sequential stages.

I. Real-Time Imbalance

Positive and negative imbalance are represented by

$I_{t,\omega}^{+} \geq P_{t,\omega}^{actual} - P_t^{DA}$	(32)
$I_{t,\omega}^{-} \geq P_t^{DA} - P_{t,\omega}^{actual}$	(33)

With

$I_{t,\omega}^{-}, I_{t,\omega}^{+} \geq 0$	(34)

The corresponding settlement cost/revenue is incorporated into (15).

J. N−1 Deliverability

For each dispatch group , the surviving flexibility after the loss of the critical component is constrained by

$F_{d,t,\omega}^{sury} \geq R_{d,t}^{comm}$	(35)

For an N−1 group,

$F_{d,t,\omega}^{sury} = \sum_{j \epsilon D_d} P_{j,t,\omega} - max_{j \epsilon D_d} P_{j,t,\omega}$	(36)

with an equivalent MILP linearization used in implementation.

Thus, a reserve or flexibility commitment cannot rely on the full capacity of a single critical resource.

K. Logical and Big-M Constraints

Binary decisions are connected to continuous operating variables using sufficiently bounded constants:

$x_t \leq M z_t$	(37)
$x_t \geq L z_t$	(38)

The implementation uses explicit physical bounds wherever possible to avoid unnecessarily large M values and numerical instability.

VI. SOLUTION METHOD AND EXPERIMENTAL DESIGN

A. Solution Procedure

The computational workflow is:

Historical Data→Calibration→Marginals→Student-t Copula→Scenario Generation→Scenario Reduction→MILP→OOS Settlement	(39)

The optimization problem is implemented in Python and solved using the open-source HiGHS MILP solver.

B. Benchmark Models

Four principal models are evaluated:

B0 — Deterministic MILP

Forecast→MILP

B1 — Gaussian stochastic MILP

Forecast Errors→Gaussian Copula→Scenarios→MILP

B2 — Student-t stochastic MILP

Forecast Errors→Student-t Copula→Scenarios→MILP

P1 — Proposed framework

Forecast→Calibration→Student-t→Tail Protection→CVaR→MILP

C. Out-of-Sample Evaluation

For each OOS period, the bidding decision is frozen before the corresponding realization becomes available.

The realized profit is calculated as

$\Pi_d^{OOS} = R_d^{DA} + R_d^{RT} + R_d^{DR} - C_d^{operation} - C_d^{imb}$	(40)

The primary metrics are:

$\overline{\Pi} = \frac{1}{D}\sum_{d=1}^{D} \Pi_d$	(41)
$CVar_{95} = \eta + \frac{1}{0.05D}\sum_{d}(l_d - \eta)^+$	(42)

together with imbalance energy, shortfall, renewable curtailment, reliability violations, runtime, and MIP gap.

D. Scenario-Count Experiment

The retained scenario counts are

$$K = 25,50,100,200$$

The selected value of K must provide an acceptable compromise between:

1. scenario fidelity;
2. economic stability;
3. tail-risk stability;
4. computational time.

No scenario count is selected solely because it produces the highest profit.

E. Sensitivity Analysis

The principal sensitivity dimensions are:

Risk coefficient

$$\lambda \in \{0,0.1,0.25,0.5,1\}$$

CVaR confidence level

$$\alpha \in \{0.90,0.95,0.975,0.99\}$$

Scenario count

$$K \in \{25,50,100,200\}$$

Additional ablation experiments remove BESS, DR, tail protection, or CVaR individually.

F. Statistical Testing

For each OOS day,

$\Delta_d^{P1,B} = \Pi_d^{P1} - \Pi_d^B$	(43)

The paired differences are evaluated using bootstrap confidence intervals. The analysis reports:

- mean difference;
- median difference;
- 95% confidence interval;
- effect size;
- empirical -value where appropriate.

This avoids treating the daily observations of competing bidding strategies as independent samples.

The formulation above describes a single-stage optimization for a given information set. However, the VPP operates in a sequential market environment where day-ahead commitments are made before intraday adjustments and balancing actions. To reflect this chronology, we implement a rolling-horizon protocol in which decisions from previous stages are fixed and only the current-stage decision variables are optimized, while all remaining flexibility is preserved for future recourse. Let $\mathcal{S} = \{1, 2, \dots, 7\}$denote the set of sequential operational stages, with stage $s$corresponding to the decision epoch (e.g., DA bid submission, intraday adjustment, balancing activation). At stage s, the optimal decision $x_s^*$is obtained by solving:

$$x_s^* = \arg\max_{x_s} Z(x_1^*, x_2^*, \dots, x_{s-1}^*, x_s)$$

subject to all constraints defined in (1)–(47), where $x_1^*, \dots, x_{s-1}^*$are the optimal decisions from previous stages and are treated as fixed parameters. This formulation ensures that the VPP's bidding strategy is dynamically consistent, avoids look-ahead bias, and respects the sequential information disclosure typical of wholesale electricity markets.

VII. RESULTS

A. Forecast and Copula Validation

The first empirical result concerns the dependence model. The Student-t copula achieved a validation AIC of −670.7362, compared with −429.9003 for the Gaussian copula. The fitted Student-t degrees of freedom was 18.5515.

The recorded OOS log-likelihood was:

Model	Validation AIC	OOS Log-Likelihood
Gaussian Copula	−429.9003	−30.708711
Student-t Copula	−670.7362	12.990350

The Student-t specification therefore provides the stronger statistical fit in the available validation experiment.

B. Temporal Scenario Validation

The validated temporal-dependence diagnostics are:

Metric	Value
Mean absolute ACF error	0.003945
Maximum absolute ACF error	0.018733

These values indicate that the validated scenario trajectories reproduce the principal temporal correlation structure with relatively small deviations.

The previously generated 5,000-scenario run with correlation MAE = 0.009763 is not reported as a final result, because the marginal-fitting error discovered subsequently invalidated that particular scenario-generation run.

C. MILP Computational Validation

The controlled solver validation produced:

Metric	**Result**
Solver	HiGHS
Horizon	24 h
Operational stages	7
Variables	1,048
Solver status	Optimal
Objective	242,085.56
Runtime	5.70 s

This experiment verifies the mathematical implementation and solver compatibility. It is explicitly treated as a formulation-validation result, rather than an OOS economic result.

D. Benchmark Results

The final empirical benchmark table must be populated only after the OOS execution:

Metric	B0	B1	B2	P1
Expected profit	—	—	—	—
Realized OOS profit	—	—	—	—
CVaR95	—	—	—	—
Imbalance	—	—	—	—
Shortfall	—	—	—	—
Curtailment	—	—	—	—
N−1 violations	—	—	—	—
Runtime	—	—	—	—
MIP gap	—	—	—	—

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E. Scenario-Count Results

	Profit	CVaR95	Imbalance	Runtime	MIP Gap
25	—	—	—	—	—
50	—	—	—	—	—
100	—	—	—	—	—
200	—	—	—	—	—

F. Risk Sensitivity

	Profit	CVaR95	Imbalance	Flexibility
0	—	—	—	—
0.10	—	—	—	—
0.25	—	—	—	—
0.50	—	—	—	—

1.00	—	—	—	—

G. CVaR Confidence-Level Sensitivity

	Profit	CVaR	Imbalance
0.90	—	—	—
0.95	—	—	—
0.975	—	—	—
0.99	—	—	—

H. Reliability Results

The final reliability evaluation will report:

$$N_{viol}, E_{short}, RLI = 1 - \frac{N_{viol}}{N_{events}} \quad (44)$$

Model	N−1 Violations	Shortfall	Reliability Index
B0	—	—	—
B1	—	—	—
B2	—	—	—
P1	—	—	—

I. Statistical Comparison

The final paired OOS comparison is structured as:

Comparison	Mean Profit Difference	95% CI	p-value	Effect
P1 vs. B0	—	—	—	—
P1 vs. B1	—	—	—	—
P1 vs. B2	—	—	—	—

VIII. Discussion

The results obtained to this stage support the methodological rationale of the proposed framework. The Student-t copula achieved a validation AIC of −670.7362, compared with −429.9003 for the Gaussian copula, with an estimated degrees-of-freedom parameter of 18.5515. The corresponding OOS log-likelihood was 12.990350 for the Student-t model and −30.708711 for the Gaussian model. These results indicate that the joint uncertainty structure is better represented by the heavier-tailed dependence model in the available validation experiment.

The temporal scenario validation further produced an ACF MAE of 0.003945 and a maximum error of 0.018733. However, the earlier 5,000-scenario experiment with correlation error 0.009763 was rejected after identifying an incorrect marginal-fitting procedure; therefore, it is not used as final empirical evidence.

The optimization framework integrates seven-stage sequential commitment, BESS operation, demand response, N−1 surviving flexibility, scenario-dependent real-time settlement, and a linear CVaR formulation. The repaired formulation was successfully validated on controlled data, including binary storage logic, real-time buy/sell exclusivity, scenario profit/loss variables, CVaR variables, and group-level N−1 constraints.

The benchmark design allows the contribution of each methodological component to be isolated. B0 represents deterministic bidding, B1 Gaussian-copula stochastic bidding, B2 Student-t stochastic bidding, and P1 combines Student-t dependence, tail protection, and CVaR.

An important methodological implication is that statistical improvement cannot automatically be interpreted as economic improvement. The final comparison must therefore use identical OOS dates, physical constraints, settlement rules, and solver settings across all models.

VIII-A. Practical Implications

The proposed information firewall prevents delivery-day observations from influencing the corresponding pre-delivery bid. This is essential for avoiding optimistic performance estimates caused by temporal data leakage. The final OOS protocol freezes the DA bid before actual PV, wind, load, and price observations are introduced into settlement.

The framework also distinguishes **economic risk** from **physical reliability**. CVaR controls downside financial exposure, whereas N−1 constraints protect surviving flexible capability. Treating these two mechanisms separately avoids interpreting a financially conservative bid as automatically being physically reliable.

VIII-B. Limitations

The principal remaining limitation is empirical: the real-data Gate 10F execution has not yet been completed because the required SMARD/calibrated/scenario files are not currently mounted in the execution environment.

Demand response is also modeled as a flexibility parameter rather than a directly observed VPP activation series, and must therefore be identified explicitly as a modeled/proxy input.

IX. Conclusion

This paper develops a calibrated dependence-aware stochastic bidding framework for virtual power plants under joint renewable-generation, load, and electricity-price uncertainty. The framework integrates calibrated marginal uncertainty, Student-t dependence modeling, tail-protected scenario reduction, sequential market commitments, BESS and demand-response flexibility, N−1 deliverability, and CVaR-based risk control within a rolling-horizon stochastic MILP.

The empirical uncertainty analysis provides evidence supporting the Student-t dependence structure, while the scenario diagnostics demonstrate preservation of temporal dependence within the validated experiment. The repaired optimization formulation has also passed controlled solver validation using the open-source HiGHS backend.

The definitive contribution of the study, however, must be established at the **decision level**. The remaining OOS experiment will determine whether improved dependence and tail representation translate into higher realized profit, lower CVaR, lower imbalance exposure, and improved reliability relative to deterministic and alternative stochastic benchmarks. The experimental protocol explicitly prohibits selecting the scenario count or reporting superiority based on favorable outcomes alone

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